



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2007

MATHEMATICS

8:45 am – 11:50 am

Tuesday 28th August 2007

Directions to Students

• Reading Time : 5 minutes	• Total Marks 120
• Working Time : 3 hours	
• Write using blue or black pen. (sketches in pencil).	• Attempt Question 1 – 10
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

This paper has been prepared independently of the Board of Studies NSW to provide additional exam preparation for students. Although references have been reproduced with permission of the Board of Studies NSW, the publication is in no way connected with or endorsed by the Board of Studies NSW.

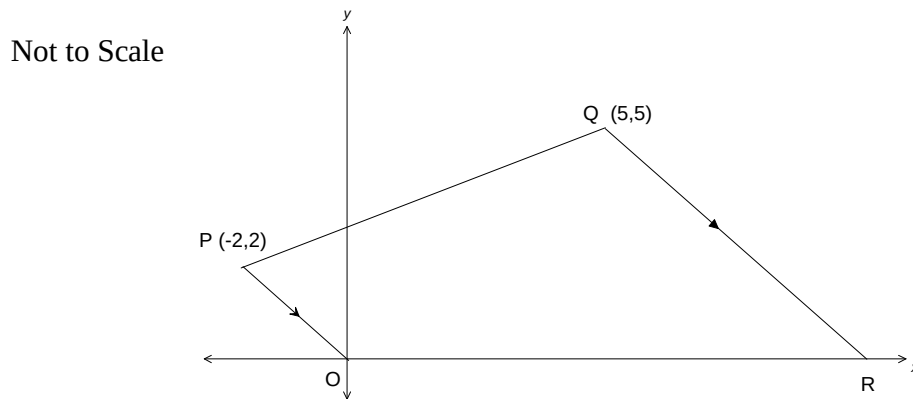
Total marks (120)
Attempt Questions 1 – 10
All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 MARKS)	Use a SEPARATE writing booklet	Marks
(a)	Evaluate $\pi^{-1.6}$ correct to 2 significant figures.	2
(b)	Write down the exact value of	
	(i) $\operatorname{cosec} \frac{\pi}{6}$	1
	(ii) $10^{\log_{10} \sqrt{2}}$	1
(c)	Paul paid Jonestown mechanical repairs \$510.00 for his car to be serviced. This price included a 15% discount from the standard cost of the service. What was the standard cost of the service.	2
(d)	Factorise fully $2 - 54x^3$.	2
(e)	Solve $ y - 2 = 6$, for y .	2
(f)	Solve $5 - \frac{a}{9} < 3$, for a .	2

QUESTION 2**(12 MARKS)**

Use a SEPARATE writing booklet

Marks**(a)**

In the diagram, OPQR is a trapezium with OP parallel to RQ . The co-ordinates of O , P and Q are $(0, 0)$, $(-2, 2)$ and $(5, 5)$ respectively.

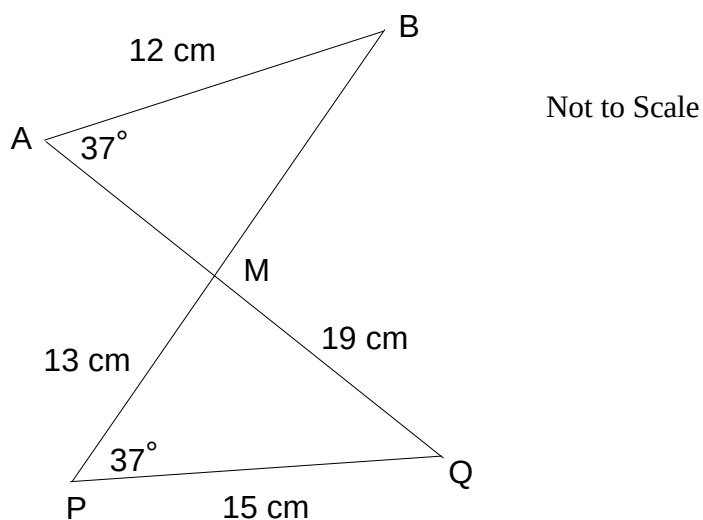
- | | | |
|-------|---|----------|
| (i) | Calculate the exact length of OP . | 1 |
| (ii) | Write down the gradient of OP . | 1 |
| (iii) | What is the size of angle POR . | 1 |
| (iv) | Find the equation of the line RQ , and hence find the co-ordinates of R . | 2 |
| (v) | Show that the perpendicular distance from O to RQ is $5\sqrt{2}$ units. | 2 |
| (vi) | Hence, or otherwise, calculate the area of the trapezium $OPQR$. | 2 |
| (b) | Shade the region in the Cartesian plane for which the inequalities $y < x - 1$, $y \geq 0$ and $x \geq 4$ hold simultaneously. | 3 |

QUESTION 3 (12 MARKS) Use a SEPARATE writing booklet

Marks

(a)

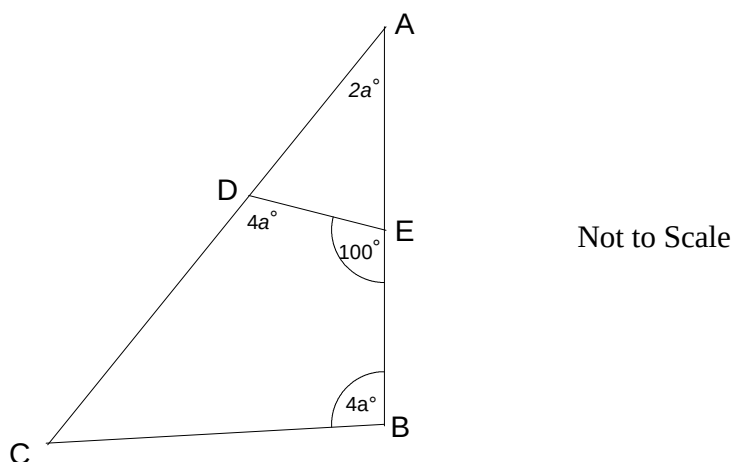
2



In the diagram, triangles BAM and QPM are similar. $BA = 12\text{ cm}$, $QP = 15\text{ cm}$, $QM = 19\text{ cm}$ and $PM = 13\text{ cm}$. Find the length of side AM .

(b)

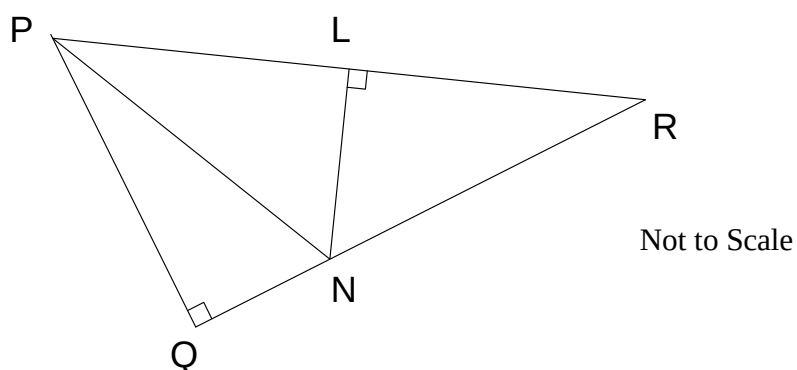
3



In the diagram ABC is a triangle and E and D are points on AB and AC respectively. Angle $CAB = 2a^\circ$, angle $EBC = \text{angle } CDE = 4a^\circ$ and angle $DEB = 100^\circ$. Find the value of a showing full working and give reasons.

QUESTION 3 (Continued)**Marks**

(c)



In the diagram PQR is a triangle with a right angle at Q. L is the midpoint of PR, and N is the point where the perpendicular to PR at L meets QR.

- (i) Show that the triangles PNL and RNL are congruent, giving reasons. 2
- (ii) Suppose that it is also given that NP bisects the angle QPR. Find
- (α) the size of angle NRL 2
- (β) the exact ratio LN : PR 2
- (d) Write down the sum of the interior angles of a regular Heptagon (7 sided polygon). 1

QUESTION 4 (12 MARKS) Use a SEPARATE writing booklet

Marks

(a) Differentiate with respect to x :

(i) $y = \log_e (3 + 2x^2)$. **1**

(ii) $y = (3e^x - 5)^7$. **2**

(iii) $y = \frac{\sin 3x}{x}$ **2**

(iv) $y = x\sqrt{x}$ **2**

(b) Find the equation of the tangent to the curve $y = -\frac{2}{x}$ at the point $(-1, 2)$. **3**

(c) Find $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 3x + 1}{2x^2 + 5} \right)$ **2**

QUESTION 5 (12 MARKS) Use a SEPARATE writing booklet **Marks**

- (a) The roots of the equation $2x^2 + 4x - 1 = 0$ are α and β .
- (i) Write down the values of $(\alpha + \beta)$ and $\alpha\beta$. **1**
- (ii) Evaluate $(\alpha^2 + \beta^2)$ **2**
- (b) Consider the quadratic equation $x^2 - (k - 2)x + (k + 1) = 0$, where 'k' is a constant.
- (i) Show that the discriminant is $(k^2 - 8k)$. **2**
- (ii) Find the values of 'k' for which the equation has real roots. **1**
- (c) A parabola has an equation in the form $x^2 - 12x = 8y - 52$
- (i) Express this equation in the form $(x - h)^2 = 4a(y - k)$ **1**
- (ii) Hence find the:
- (α) co-ordinates of the vertex **1**
- (β) co-ordinates of the focus **1**
- (γ) equation of the directrix **1**
- (d) Solve the equation $(5^x)^2 + 5^x - 2 = 0$, for x . **2**

QUESTION 6

(12 MARKS)

Use a SEPARATE writing booklet

Marks

(a) Find

$$(i) \int \frac{1}{4x^4} dx \quad 1$$

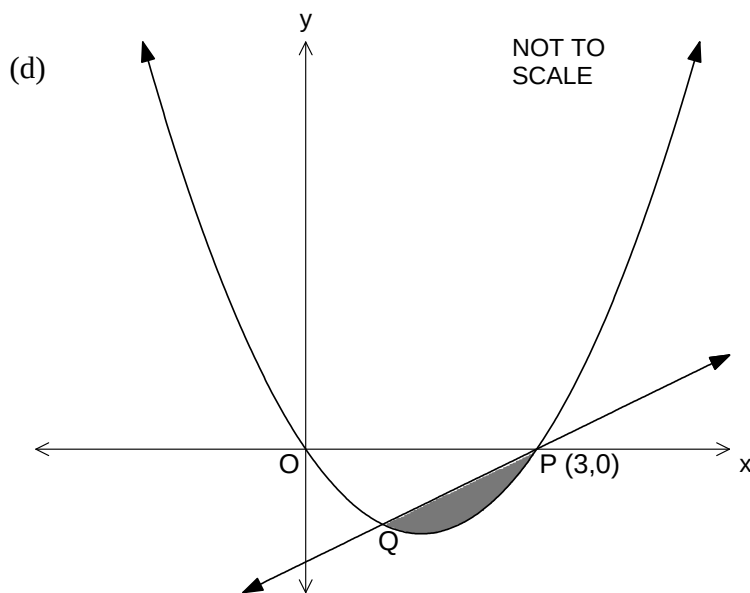
$$(ii) \int \frac{2x}{x^2+10} dx \quad 1$$

$$(b) \text{ Evaluate } \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x \, dx \quad 2$$

(c) Use Simpson's Rule with three function values to find an approximation for **3**

$$\int_3^7 \frac{x}{\ln x} dx.$$

Give your answer correct to one decimal place.



The graphs of $y = x - 3$ and $y = x^2 - 3x$ intersect at the points P (3,0) and Q, as shown in the diagram.

(i) Find the co-ordinates of Q. **2**(ii) Find the area of the shaded region bounded by $y = x^2 - 3x$ and $y = x - 3$. **3**

QUESTION 7

(12 MARKS) Use a SEPARATE writing booklet

Marks

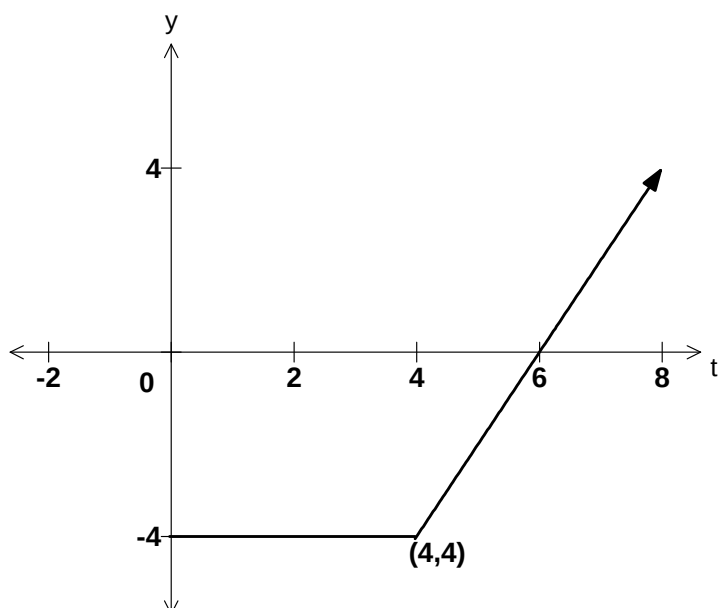
(a) Consider the curve $y = x^3 - 3x^2 + 3x - 1$.

(i) Show that the curve has only one stationary point, find its co-ordinates and determine its nature. **3**

(ii) State the values of x for which the curve is concave up. **1**

(iii) State the values of x for which the curve is increasing. **1**

(b)

**2**

The graph shows the function $y = f(x)$ whose domain is $x \geq 0$.

Trace or copy this graph onto your answer booklet

On the same axes, sketch the graph of the function $y = f'(x)$.

(c) The curve $y = f(x)$, $0 \leq x \leq 2\pi$ has a stationary point at $x = \frac{\pi}{4}$ and $f''(x) = \cos x + \sin x$. **3**

Find the x co-ordinate at which there is another stationary point.

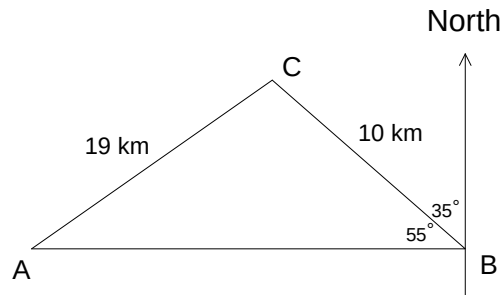
(d) If $y = x\sqrt{x+1}$, show that $\frac{dy}{dx} = \frac{3x+2}{2\sqrt{x+1}}$. **2**

QUESTION 8

(12 MARKS) Use a SEPARATE writing booklet

Marks

(a)



In the diagram, the point B is due east of point A. The point C is 19 km from point A and 10 km from point B. Point C is North 35° West of point B. Find the true bearing of point C from point A.

3

(b) The velocity of a particle, moving in a straight line, is given by

$$\frac{dx}{dt} = 2 - 4 \sin t \text{ for } 0 \leq t \leq 2\pi$$

where $\frac{dx}{dt}$ is measured in metres per second, t is measured in seconds and x in metres.

- (i) At what times during this period is the particle at rest. **2**
- (ii) Sketch the graph of velocity $\frac{dx}{dt}$ as a function of time for $0 \leq t \leq 2\pi$. **2**
(This sketch should be approximately half of one page)
- (iii) What is the maximum velocity of the particle during this period? **2**
- (iv) Calculate the exact total distance travelled by the particle **3**
between $t=0$ and $t=\frac{\pi}{2}$

QUESTION 9

(12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The population P of ants in a colony is determined by $P = 500 \times e^{\frac{t}{4}}$ where t is the time in weeks since the colony was originally established.
- (i) Find the size of the colony after 10 weeks. **1**
- (ii) How long would it take for the population to reach 10 000? **1**
- (iii) Find $\frac{dP}{dt}$ and explain what this means for the population trend? **2**
- (iv) Sketch the graph of $\frac{dP}{dt}$ against t . **2**
- (b) Paul and Wendy borrow \$20000 from the Miami Bank. This loan plus interest is to be repaid in equal monthly instalments of \$399 over five years. Interest of 7.2% p.a is compounded monthly on the balance owing at the start of each month.
- Let A_n be the amount owing after n months.
- (i) Over the five year repayment period, how much interest is charged? **1**
- (ii) Show that $A_1 = 19721$ **1**
- (iii) Clearly show that $A_2 = 20000 \times 1.006^2 - 399(1 + 1.006)$. **1**
- (iv) Deduce then that $A_n = 66500 - 46500 \times 1.006^n$ **2**
- (v) After two years of repayments Paul and Wendy decide on the very next day to repay the loan in one full payment. **1**
How much will this one payment be?

QUESTION 10 (12 MARKS) Use a SEPARATE writing booklet

Marks

- (a) The area bounded by the curve $y = \frac{1}{x}$, the x-axis and the ordinates $x = a$ and $x = 4$ is rotated about the x-axis. If the volume generated is $\frac{\pi}{2} \text{ unit}^3$, where $0 < a < 4$, find the value of a . **3**
- (b) If $x \sec \theta = y \tan \theta$, prove that $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$ **3**
- (c) If the straight line $y = mx$ is a tangent to the curve $y = e^{\frac{x}{2}}$, find the exact value of m . Clearly show your working. **3**
- (d) The sum of the first 'n' terms of a series is given by $S_n = n^2 + 6n$. **3**
Prove that this series is arithmetic.

(Note: It will be insufficient to outline a proof which involves evaluating $S_1, S_2, \text{etc.}, T_1, T_2, \text{etc.}$)

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2007

MATHEMATICS

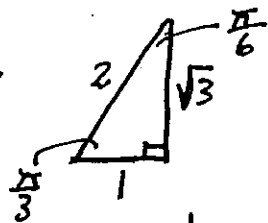
- This is a study resource for HSC preparation.
- This document is designed to help students understand the questions.
- The solutions should be treated as aids only.
- There may be better solutions to some questions.

Question 1

(a) $\pi^{-1.6} \approx 0.16$

From calculator
0.160162265...

(b) (i) $\operatorname{cosec} \frac{\pi}{6} = \frac{2}{1} = 2$



(ii) $10^{\log_{10} \sqrt{2}} = \sqrt{2}$ Rule: $a^{\log_a x} = x$

(a) IAW - Correct calc^e

IAW - Correct rounding

(i) IAW - Correct answer

(ii) IAW - Correct answer

(c) 85% of cost of standard service = \$510.00

1% of cost of standard service = $\$ \frac{510.00}{85}$

Cost of standard service = $\$ \left(\frac{510.00 \times 100}{85} \right)$

$= \$600.00$

IAW - Some attempt at unitary method

IAW - Correct answer

(d) $2 - 54x^3 = 2(1 - 27x^3)$

$= 2(1 - 3x)(1 + 3x + 9x^2)$

IAW - take out a factor of 2

IAW - correct answer

(e) $|y - 2| = 6$

$\pm(y - 2) = 6$

$y - 2 = 6$ or $y - 2 = -6$

$y = 8$ or $y = -4$

(e) Well done by most

IAW - $y = 8$ IAW - $y = -4$

(f) $5 - \frac{a}{9} < 3$

$-\frac{a}{9} < -2$

$a > 18$

(f) Many careless errors made here

IAW - Reasonable attempt

IAW - Correct answer

Question 2

(i) $O(0,0); P(-2,2)$

$$OP^2 = 2^2 + 2^2$$

$$OP^2 = 4 + 4$$

$$OP^2 = 8$$

$$OP = \sqrt{8} = 2\sqrt{2}$$

(i) Well answered

(ii) $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m = \frac{2 - 0}{-2 - 0}$$

$$m = -1$$

(ii) well answered

(iii) $\tan \theta = m$

$$\tan \widehat{POR} = -1$$

$$\widehat{POR} = 135^\circ$$

(iii) many students did not know this result !!

(iv) $QR \parallel PO$

$$m_{QR} = m_{PO} = -1$$

Equation of QR:

$$y - y_1 = m(x - x_1)$$

using $A(5,5)$

$$y - 5 = -1(x - 5)$$

$$y - 5 = -x + 5$$

$$x + y - 10 = 0$$

Now let $y = 0$ for co-ordinates of R

$$\therefore x = 10$$

point $R = (10,0)$

(iv) many students did not read the second part of this question ----
→ Find the co-ord. of R

IAW - Equation of Line

IAW - Co-ord. of R

$$(v) \quad d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$BC: x + y - 10 = 0$$

Point $O(0,0)$

$$d = \frac{|1 \times 0 + 1 \times 0 - 10|}{\sqrt{1^2 + 1^2}}$$

$$d = \frac{10}{\sqrt{2}}$$

$$d = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$d = 5\sqrt{2} \text{ units}$$

(v) Generally well answered

IAW - Substitution into Formula

IAW - Rationalising the denominator.

(vi) Distance QR.

$$\overline{QR}^2 = (5-10)^2 + (5-0)$$

$$\overline{QR}^2 = (-5)^2 + 5^2$$

$$\overline{QR}^2 = 25 + 25$$

$$\overline{QR}^2 = 50$$

$$\overline{QR} = \sqrt{50} = 5\sqrt{2}$$

(vi) Many students had difficulty with this

IAW - Distance QR

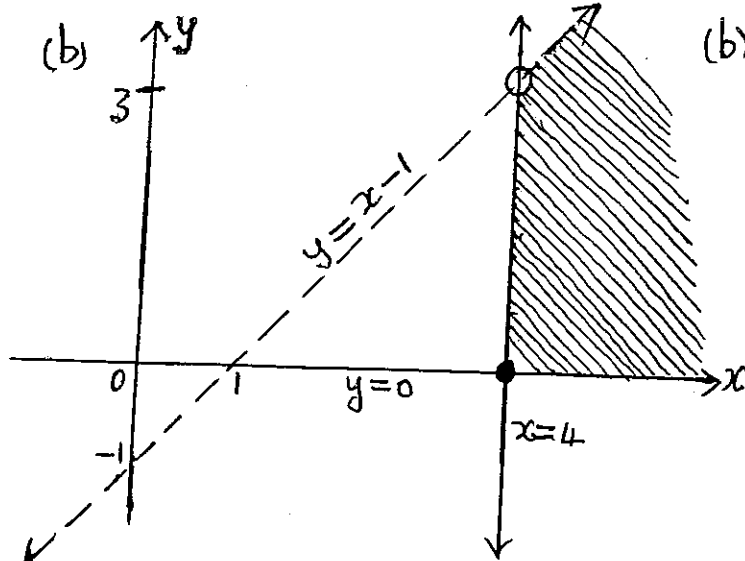
$$\text{Trapezium: } A = \frac{h}{2} (a+b)$$

$$A = \frac{5\sqrt{2}}{2} (2\sqrt{2} + 5\sqrt{2})$$

$$A = \frac{5\sqrt{2}}{2} \times 7\sqrt{2}$$

$$A = 35 \text{ units}^2$$

IAW - Use of Trapezium Formula



(b) Poorly answered by many

Marks were deducted for

1. Not showing intercepts
2. Not labelling diagram
3. Not showing dotted line for $y = x - 1$

Note - use a ruler

- use a pencil. 3

Question 3

(a) $\frac{AM}{13} = \frac{12}{15}$

$AM = \frac{12 \times 13}{15}$

$AM = 10.4 \text{ cm}$

(Note: Ratios of corresponding sides are equal)

(b) $\widehat{AED} = 180^\circ - 100^\circ = 80^\circ$

(Angle sum at E on a straight line AB is 180°)

$2a^\circ + 80^\circ = 6a^\circ$. External angle of $\triangle ADE$ equals the sum of the interior opposite angles.

$\therefore 4a = 80$

$a = 20$

Note There is more than one approach here.

(c)

(i) In the Δ 's PNL and RNL

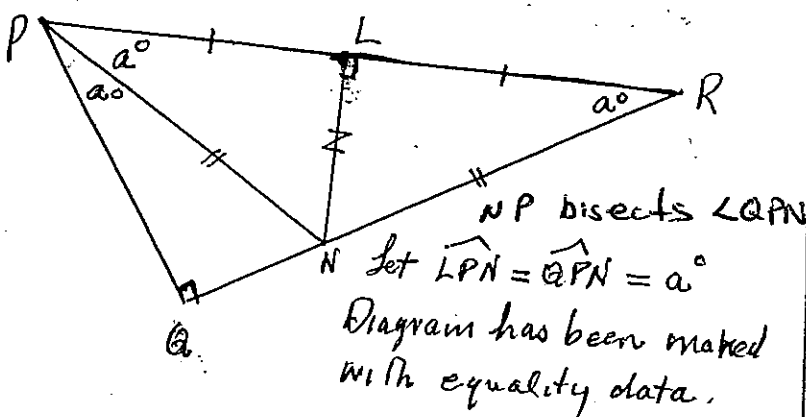
NL is a common side

$\widehat{NLP} = \widehat{NLR} = 90^\circ$ ($NL \perp PR$ given)

$LP = LR$ (L is the midpoint of PR , given)

$\therefore \triangle PNL \equiv \triangle RNL$ (SAS)

(ii)



Well done

MANY DIFFERENT OPTIONS USED. QUITE WELL DONE.

$a = 40$ WITHOUT SOME CORRECT SETTING OUT WAS PENALISED

1 MK AWARDED FOR FINDING ALL THREE PARTS OF PROOF

FOR CORRECT CONGRUENCY RULE

MANY HAD NO REASON OFFERED FOR $\angle QPN = \angle NPL$.

(11) continued.

(a) In $\triangle PQR$ $2a^\circ + a^\circ + 90^\circ = 180^\circ$

$$3a = 90$$

$$a = 30$$

$$\therefore \widehat{NRL} = 30^\circ$$

(b) $\frac{LN}{PL} = \tan 30^\circ = \frac{1}{\sqrt{3}}$

Now since $PL = \frac{1}{2} PR$,

we have: $\frac{LN}{\frac{1}{2} PR} = \frac{1}{\sqrt{3}}$

$$\therefore \frac{LN}{PR} = \frac{1}{2\sqrt{3}}$$

(d) Formula: $(2n-4)$ right-angles

$$\begin{aligned} \text{Angle sum} &= (14-4) \times 90^\circ \\ &= 900^\circ \end{aligned}$$

AGAIN, A NUMBER OF
CORRECT OPTIONS USED.
MOST SUCCESSFULLY.

MANY HAD GOOD SOLN'S
TO THIS.

SOME MADE A SOLID
START BUT COULD NOT
COMPLETE. PART MKS
AWARDED

Well done.

Question 4

(a) (i) $y = \log_e(3 + 2x^2)$

$$\frac{dy}{dx} = \frac{4x}{3+2x^2}$$

(ii) $y = (3e^x - 5)^7$

$$\begin{aligned}\frac{dy}{dx} &= 7(3e^x - 5)^6 \times 3e^x \\ &= 21e^x(3e^x - 5)^6\end{aligned}$$

(iii) $y = \frac{\sin 3x}{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{x \times 3 \cos 3x - (\sin 3x) \times 1}{x^2} \\ &= \frac{3x \cos 3x - \sin 3x}{x^2}\end{aligned}$$

(iv) $y = x\sqrt{x} = x \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3}{2} x^{\frac{1}{2}} \\ &= \frac{3\sqrt{x}}{2}\end{aligned}$$

(b) $y = -2x^{-1}$, point $(-1, 2)$

$$\frac{dy}{dx} = 2x^{-2} = \frac{2}{x^2}$$

at $x = -1$, $m = \frac{2}{(-1)^2} = 2$

tangent: $y - y_1 = m(x - x_1)$

$$y - 2 = 2(x + 1)$$

$$y - 2 = 2x + 2$$

$$2x - y + 4 = 0$$

Marks

many forgot formula.

many forgot to differentiate $3e^x$.for finding u' and v'

correct arrangement in formula.

changing to index form ie $x^{\frac{1}{2}}$ differentiating correctly
many used product rule
but this was the long way round.

differentiating correctly

gradient of tangent

for equation of line

alternative: $y = 2x + 4$

$$(c) \lim_{x \rightarrow \infty} \left(\frac{x^2 - 3x + 1}{2x^2 + 5} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1 - \frac{3}{x} + \frac{1}{x^2}}{2 + \frac{5}{x^2}} \right)$$

$$= \frac{1 - 0 + 0}{2 + 0}$$

$$= \frac{1}{2}$$

1 dividing through by the highest power of x
ie x^2

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} - \frac{3x}{x^2} + \frac{1}{x^2}}{\frac{2x^2}{x^2} + \frac{5}{x^2}}$$

1 correct answer.

Question 5

(a) $2x^2 + 4x - 1 = 0$

$a = 2, b = 4, c = -1$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{4}{2} = -2$

$\alpha\beta = \frac{c}{a} = -\frac{1}{2}$

} Both right ✓

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (-2)^2 - 2(-\frac{1}{2})$

$= 4 + 1$

$= 5$

(b) $x^2 - (k-2)x + (k+1) = 0$

(i) $\Delta = b^2 - 4ac$

$\Delta = [-(k-2)]^2 - 4(1)(k+1)$

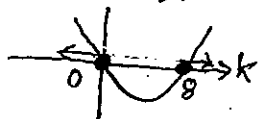
$\Delta = k^2 - 4k + 4 - 4k - 4$

$\Delta = k^2 - 8k$

(ii) $\Delta \geq 0$ for real roots

$k^2 - 8k \geq 0$

$k(k-8) \geq 0$

Ans $k \leq 0$ or $k \geq 8$

Not many problems in this Q.

$$\begin{aligned}
 (c) \quad x^2 - 12x &= 8y - 52 \\
 x^2 - 12x + (-6)^2 &= 8y - 52 + 36 \\
 (x-6)^2 &= 8y - 16 \\
 (x-6)^2 &= 8(y-2) \\
 (x-6)^2 &= 4(2)(y-2)
 \end{aligned}$$

$$(d) \text{ Vertex : } (6, 2)$$

$$\text{Focal length} = 2.$$

$$(\beta) \text{ Focus : } (6, 4)$$

$$(\gamma) \text{ Directrix : } y = 0$$

* If brackets () were not used for coordinates 1 mark was deducted in (d) and (β)

$$(d) (5^x)^2 + 5^x - 2 = 0$$

$$\text{Let } t = 5^x$$

$$t^2 + t - 2 = 0$$

$$(t+2)(t-1) = 0$$

$$t = -2 \text{ or } t = 1$$

$$\therefore 5^x = -2 \text{ N/A as } 5^x > 0 \text{ for all } x$$

$$5^x = 1$$

$$5^x = 5^0$$

$$\underline{x = 0}$$

Question 6

(a) (i) $\int \frac{1}{4x^4} dx$
 $= \int \frac{1}{4} x^{-4} dx$
 $= \frac{1}{4} \frac{x^{-3}}{-3} + c$
 $= -\frac{1}{12x^3} + c$

✓

(ii) $\int \frac{2x}{x^2+10} dx$
 $= \ln(x^2+10) + c$

✓

(b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x dx$
 $= \left[\tan x \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$
 $= \tan \frac{\pi}{4} - \tan \frac{\pi}{6}$
 $= 1 - \frac{1}{\sqrt{3}}$
 $= \frac{\sqrt{3}-1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}(\sqrt{3}-1)}{3} \text{ or } \frac{3-\sqrt{3}}{3}$

✓

✓

✓

Many incorrectly dealt with this question as a logarithm.

Two issues to deal with...

(1) $\frac{1}{4}$

(2) x^{-4}

answered well.

$\int_a^b \sec^2 x = [\tan x]_a^b$

exact value leave in this form.

completing the table of values

x	3	5	7
$\frac{x}{\ln x}$	$\frac{3}{\ln 3}$	$\frac{5}{\ln 5}$	$\frac{7}{\ln 7}$
	y_1	y_2	y_3

$\Gamma \equiv \frac{b-a}{n} [(y_1+y_3) + 4y_2]$

$\Gamma \equiv \left(\frac{7-3}{6}\right) [(2.7307 + 3.5973) + 4(3.1067)]$

$\Gamma \equiv 12.5 \text{ c.i.d.p}$

✓

✓ correctly substituting into formula

NOTE: $\frac{b}{3} = \frac{2}{3}$

$$(d) (i) \quad y = x - 3 \quad \text{--- (1)}$$

$$y = x^2 - 3x \quad \text{--- (2)}$$

$$(2) - (1): x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1 \text{ or } 3.$$

$$\text{when } x = 1 \quad y = -2$$

$\therefore$ point A is (1, -2)

✓

✓

$$(ii) \quad A = \int_1^3 (x^2 - 3x) dx - \int_1^3 (x - 3) dx$$

$$A = \int_1^3 (x^2 - 4x + 3) dx$$

✓

$$A = \left[\frac{x^3}{3} - 2x^2 + 3x \right]_1^3$$

✓

$$A = \left[(9 - 18 + 9) - \left(\frac{1}{3} - 2 + 3 \right) \right]$$

$$A = \frac{4}{3} \text{ units}^2$$

✓

Area between two curves
is best answered using
the formula: $\int_a^b f(x) - g(x) dx$.

Note: 1 mark deducted if you
had the area as $-\frac{4}{3} \text{ units}^2$.

Question 7

(a) $y = x^3 - 3x^2 + 3x - 1$

$\frac{dy}{dx} = 3x^2 - 6x + 3$

$\frac{d^2y}{dx^2} = 6x - 6$

(i) Let $\frac{dy}{dx} = 0$

$3x^2 - 6x + 3 = 0$

$3(x^2 - 2x + 1) = 0$

$(x-1)^2 = 0$

$x = 1$

hence only one stationary point.

When $x = 1$ $y = 1 - 3 + 3 - 1 = 0$

$\frac{d^2y}{dx^2} = 6(1) - 6 = 0$

This will be a horizontal point of inflection at $(1, 0)$

Concavity Test:

x	0	1	2
$\frac{d^2y}{dx^2}$	-6	0	6

ie $\frac{d^2y}{dx^2} \neq 0$

This confirms inflection.

(ii) Concave up: $f''(x) > 0$

$6x - 6 > 0$

$x > 1$

(iii) $f'(x) > 0$
 $(x-1)^2 > 0$

all real x , except $x \neq 1$.

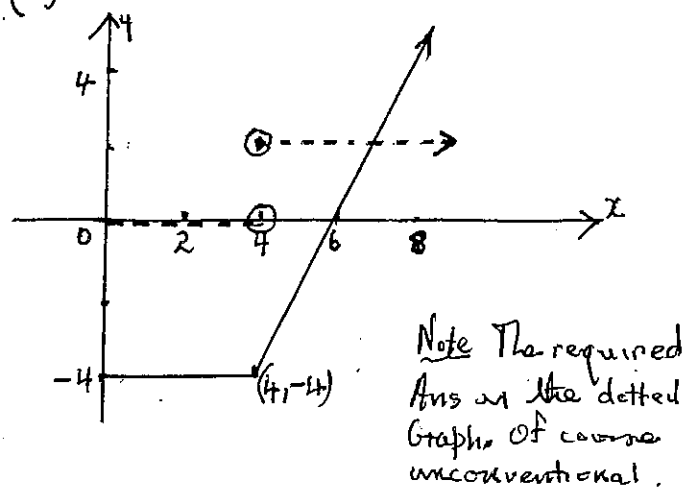
✓ finding the x -value of the S.P

✓ y -value of S.P

✓ determining nature of S.P

← marks weren't awarded for the concavity test but this should be shown.

(b)



✓ lines $y=2$ or $y=1$ and

✓ discontinuity at $x=4$

(c) $f''(x) = \cos x + \sin x$

$$f'(x) = \sin x - \cos x + C$$

Now when $f'(0) = 0$, $x = \frac{\pi}{4} \therefore C = 0$

$$\therefore f'(x) = \sin x - \cos x$$

For stationary points $f'(x) = 0$

$$\sin x - \cos x = 0$$

$$\sin x = \cos x$$

$$\tan x = 1 > 0$$

In the stated domain

x in 1st and 3rd quadrant

Basic acute angle $= \frac{\pi}{4}$

Hence the other stationary point is at $x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

(d) $y = x\sqrt{x+1} = x(x+1)^{\frac{1}{2}}$

$$\frac{dy}{dx} = 1 \times (x+1)^{\frac{1}{2}} + \frac{1}{2}(x+1)^{-\frac{1}{2}} \times 1 \times x$$

$$= \sqrt{x+1} + \frac{x}{2\sqrt{x+1}}$$

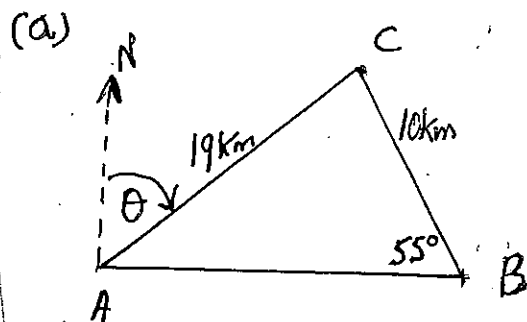
$$= \frac{2(x+1) + x}{2\sqrt{x+1}}$$

$$= \frac{3x+2}{2\sqrt{x+1}}$$

✓ correct application of product rule.

✓ simplification.

Question 8



using The sine rule :

$$\frac{\sin \hat{BAC}}{10} = \frac{\sin 55^\circ}{19}$$

$$\sin \hat{BAC} = \frac{10 \sin 55^\circ}{19}$$

$$\hat{BAC} = 25^\circ 32'$$

True bearing of C from A :

$$\theta = 90^\circ - 25^\circ 32' = 64^\circ 28'$$

ie T 064°28'

(b) $v = \frac{dx}{dt} = 2 - 4 \sin t$

(1) The particle is at rest when $v = 0$

$$2 - 4 \sin t = 0$$

$$\sin t = \frac{1}{2}$$

Now considering the domain $0 \leq t \leq 2\pi$

$\sin t > 0 \therefore t$ in 1st, 2nd. quads

Basic acute angle = $\frac{\pi}{6}$

$\therefore$ Particle at rest when $t = \frac{\pi}{6}$ sec

and when $t = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ sec

True bearing is written in 3 digits.

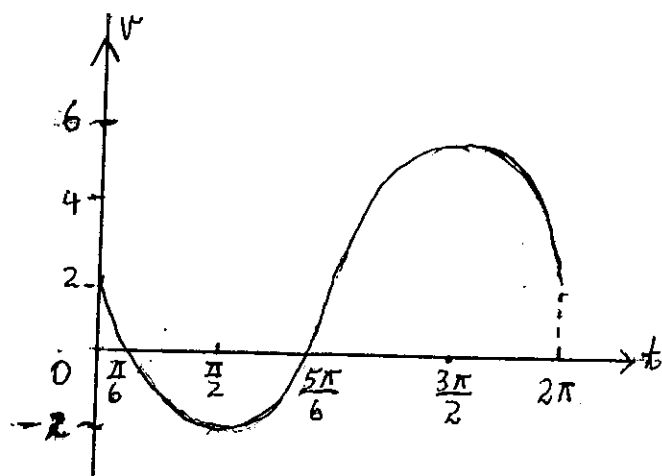
Many missed out on that (Did not penalise this time)

Bonus $\frac{\pi}{6}$, $\frac{5\pi}{6}$ had to be there for 1 mark.

1 right No mark

more than 2 answers no mark

(ii)



Points of intersection
have to be shown on the
t-axis ie $\frac{\pi}{6}$, $\frac{5\pi}{6}$

✓

- (iii) Maximum velocity occurs
when $\sin t$ is a minimum
obviously $-1 \leq \sin t \leq 1$
This occurs at $t = \frac{3\pi}{2}$ ie $\sin t = -1$
 $\therefore$ Max Velocity $= 2 - 4(-1)$
 $= 6 \text{ m/sec}$

Answers from graph
accepted for 2 marks.

- (iv) Now between $t=0$ and $t=\frac{\pi}{2}$.
The particle changes direction
at $t = \frac{\pi}{6}$ (at $v=0$)

$$\begin{aligned} \therefore \text{Total distance travelled} &= \int_0^{\frac{\pi}{6}} (2-4\sin t) dt + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2-4\sin t) dt \\ &= \left[2t + 4\cos t \right]_0^{\frac{\pi}{6}} + \left[2t + 4\cos t \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left| \left(\frac{\pi}{3} + 4\cos \frac{\pi}{6} \right) - (0 + 4\cos 0) \right| + \left| \left(\pi + 4\cos \frac{\pi}{2} \right) - \left(\frac{\pi}{3} + 4\cos \frac{\pi}{6} \right) \right| \\ &= \left| \frac{\pi}{3} + 4 \times \frac{\sqrt{3}}{2} - 4 \right| + \left| \pi + 4(0) - \frac{\pi}{3} - 4 \times \frac{\sqrt{3}}{2} \right| \\ &= \left| \frac{\pi}{3} + 2\sqrt{3} - 4 \right| + \left| \pi - \frac{\pi}{3} - 2\sqrt{3} \right| \\ &= \frac{\pi}{3} + 2\sqrt{3} - 4 + \left| \frac{2\pi}{3} - 2\sqrt{3} \right| \\ &= \frac{\pi}{3} + 2\sqrt{3} - 4 + \frac{2\pi}{3} + 2\sqrt{3} = \left(\frac{\pi}{3} + 4\sqrt{3} - 4 \right) \text{m} \end{aligned}$$

very few reached the
appropriate end.
Anything close to 3rd
step was awarded
a mark.

Question 9

(a) $P = 500 \times e^{\frac{t}{4}}$

(i) Let $t = 10$

$$P = 500 \times e^{\frac{10}{4}}$$

$$P = 500 \times e^{\frac{5}{2}}$$

$$P \approx 6091$$

∴ After 10 weeks colony has 6091 ants

(ii) $t = ?$ when $P = 10000$

$$500 \times e^{\frac{t}{4}} = 10000$$

$$e^{\frac{t}{4}} = 20$$

$$\frac{t}{4} = \ln 20$$

$$t = 4 \ln 20$$

$$t \approx 11.98$$

∴ Colony would take 11.98 weeks to reach 10,000.

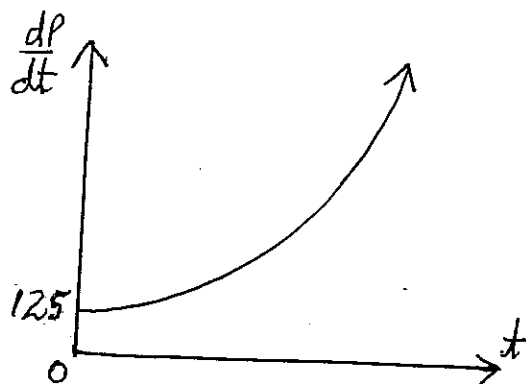
(iii) $\frac{dP}{dt} = \frac{1}{4} \times 500 \times e^{\frac{t}{4}}$

$$= 125 \times e^{\frac{t}{4}}$$

As $\frac{dP}{dt} > 0$ for all t , This means

That the population will always be increasing.

(iv)



generally well done.

1 correct answer

generally well done.

1 correct answer

1 finding $\frac{dP}{dt}$

1 explaining that population is increasing

2 correct shape, vertical intercept + domain.

(b) Loan Amount = \$20000
Monthly Instalment \$399 over 5 yrs
Interest 7.2% p.a compounded monthly.

$$(i) \text{ Interest} = (\$399) \times (12 \times 5) - \$20000 \\ = \$3940$$

$$(ii) \text{ Using } A = PR^n \text{ where } R = 1 + \frac{r}{100} \\ A_1 = 20000(1.006)^1 - 399 = \$19721$$

$$(iii) A_2 = A_1(1.006)^1 - 399$$

$$A_2 = [20000(1.006) - 399]1.006 - 399$$

$$A_2 = 20000(1.006)^2 - 399(1 + 1.006)$$

(iv) continuing from part (iii) and following the pattern, apparent:
Then after 'n' payments the amount A_n owing is:

$$A_n = 20000(1.006)^n - 399(1 + 1.006 + \dots + 1.006^{n-1})$$

↑
This is a GP with n terms, $a = 1$, $r = 1.006$
and using $SA = \frac{a(r^n - 1)}{r - 1}$

$$\therefore A_n = 20000(1.006)^n - 399 \left[\frac{1(1.006^n - 1)}{1.006 - 1} \right]$$

$$A_n = 20000(1.006)^n - 66500(1.006^n - 1)$$

$$A_n = 20000(1.006)^n - 66500(1.006^n) + 66500$$

$$A_n = 66500 - 46500(1.006)^n$$

(v) The One payment = A_{24} .

$$A_{24} = 66500 - 46500(1.006)^{24}$$

$$A_{24} = \$12820.99$$

$$\text{or } \$12821.$$

Simplest part of Q9 worst done.

had to show
 $20000 \times 1.006 - 399$

had to show this step or equivalent.

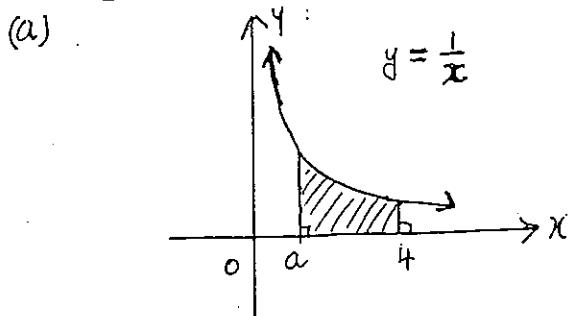
showing the expression for A_n

many people faked from here to answer.

must clearly show arithmetic.

1. many forgot that
24 months = 2 yrs.
Simple substitution into formula was given.

Question 10



$$V = \pi \int_a^b y^2 dx$$

$$\frac{\pi}{2} = \pi \int_a^4 \frac{1}{x^2} dx$$

$$\frac{1}{2} = \int_a^4 x^{-2} dx$$

$$\frac{1}{2} = \left[\frac{x^{-1}}{-1} \right]_a^4$$

$$\frac{1}{2} = - \left[\frac{1}{x} \right]_a^4$$

$$\frac{1}{2} = - \left[\frac{1}{4} - \frac{1}{a} \right]$$

$$\frac{1}{2} = \frac{1}{a} - \frac{1}{4}$$

$$\frac{1}{a} = \frac{3}{4}$$

$$a = \frac{4}{3} = 1\frac{1}{3}$$

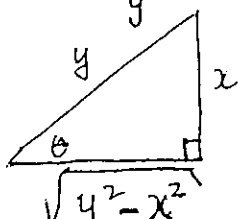
(b)

$$x \sec \theta = y \tan \theta$$

$$\frac{x}{\cos \theta} = \frac{y \sin \theta}{\cos \theta}$$

$$x = y \sin \theta$$

$$\sin \theta = \frac{x}{y}$$



⊥ For correct integral.

Too many tried to use logs.

⊥ For correct substitution

⊥ For correct answer if all else correct

⊥ For correct discovery of $x = y \sin \theta$

⊥ For discovery at $\triangle$.

Now Prove $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$
using data from The triangle

$$\text{LHS} = \tan \theta \sec \theta$$

$$= \frac{x}{\sqrt{y^2 - x^2}} \times \frac{y}{\sqrt{y^2 - x^2}}$$

$$= \frac{xy}{\sqrt{y^2 - x^2}}$$

$$= \text{RHS.}$$

(c) $y = mx$, $y = e^{\frac{x}{2}}$

Let (x_0, y_0) be the point of tangency.

$$y = e^{\frac{x}{2}}$$

$$y' = \frac{1}{2} e^{\frac{x}{2}}$$

$$\therefore m = \frac{1}{2} e^{\frac{x_0}{2}} \quad \text{--- (1)}$$

Solving $y = mx$ and $y = e^{\frac{x}{2}}$

$$mx = e^{\frac{x}{2}}$$

$$\text{So } mx_0 = e^{\frac{x_0}{2}}$$

$$\text{Hence } \frac{1}{2} e^{\frac{x_0}{2}} \times x_0 = e^{\frac{x_0}{2}} \quad \text{from (1)}$$

$$\frac{1}{2} x_0 = 1$$

$$x_0 = 2$$

$$\therefore m = \frac{1}{2} e^{2 \times \frac{1}{2}}$$

$$m = \frac{e}{2}$$

Many methods tried.
Credit given for CORRECT
Substitution.

Too many assumed
 $\tan \theta \sec \theta = \frac{xy}{y^2 - x^2}$.

1mk for correct subs
& evaluation.

1 For correct derivative

1 For attempt to solve
two equations simultaneously

1 Correct evaluation at
M.

(d) $S_n = n^2 + 6n$

$$T_n = S_n - S_{n-1}$$

$$= n^2 + 6n - [(n-1)^2 + 6(n-1)]$$

$$= n^2 + 6n - [n^2 - 2n + 1 + 6n - 6]$$

$$= n^2 + 6n - n^2 + 2n - 1 - 6n + 6$$

$$= 2n + 5$$

$$T_n - T_{n-1} = 2n + 5 - [2(n-1) + 5]$$

$$= 2n + 5 - [2n - 2 + 5]$$

$$T_n - T_{n-1} = 2n + 5 - 2n - 2 + 5$$

$$T_n - T_{n-1} = 2 \quad \text{thus 'any term'}$$

minus the term before it is constant

Hence an AP.

1 $T_n = S_n - S_{n-1}$

Most assumed an AP
existed.

1 $T_n = 2n + 5$

Many tried to find T_1, T_2, T_3
etc

1 $T_n - T_{n-1} = d$
 $d = 2$.